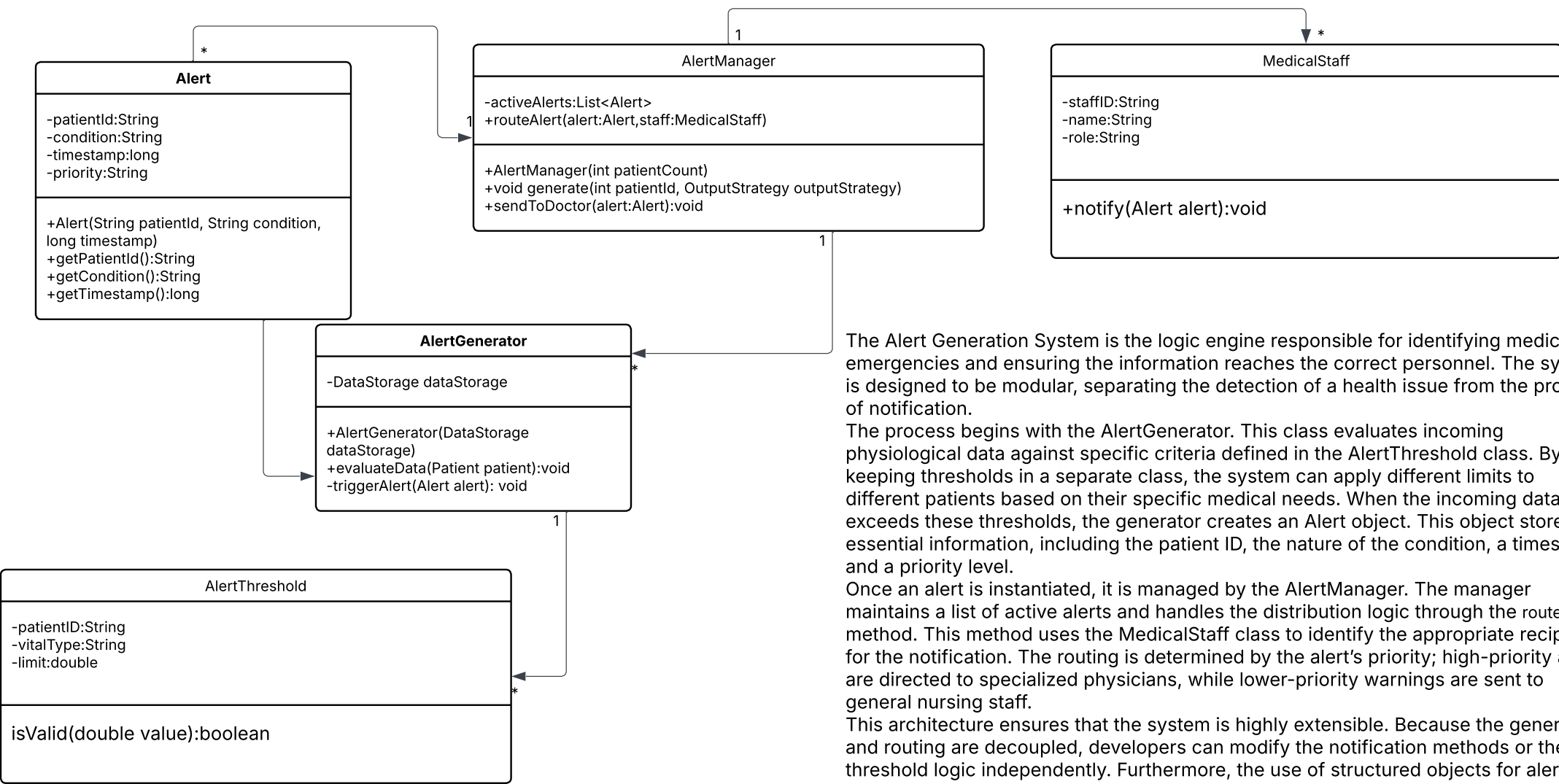




The Alert Generation System is the logic engine responsible for identifying medical emergencies and ensuring the information reaches the correct personnel. The system is designed to be modular, separating the detection of a health issue from the process of notification.

The process begins with the **AlertGenerator**. This class evaluates incoming physiological data against specific criteria defined in the **AlertThreshold** class. By keeping thresholds in a separate class, the system can apply different limits to different patients based on their specific medical needs. When the incoming data exceeds these thresholds, the generator creates an **Alert** object. This object stores essential information, including the patient ID, the nature of the condition, a timestamp, and a priority level.

Once an alert is instantiated, it is managed by the **AlertManager**. The manager maintains a list of active alerts and handles the distribution logic through the `routeAlert` method. This method uses the **MedicalStaff** class to identify the appropriate recipient for the notification. The routing is determined by the alert's priority; high-priority alerts are directed to specialized physicians, while lower-priority warnings are sent to general nursing staff.

This architecture ensures that the system is highly extensible. Because the generation and routing are decoupled, developers can modify the notification methods or the threshold logic independently. Furthermore, the use of structured objects for alerts and staff ensures that every event is documented with a clear audit trail, maintaining high standards for data integrity and clinical accountability.